

Tuhin Garai

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PROFILE

DevOps and Cloud Engineer with hands-on experience building serverless and containerized applications on AWS. Skilled in Linux administration, CI/CD automation, cloud infrastructure, and troubleshooting distributed systems, with practical exposure to Kubernetes, monitoring, and infrastructure automation.

SKILLS

DevOps & Cloud: Docker, Docker Compose, Kubernetes, Jenkins, CI/CD, AWS (EC2, VPC, S3, IAM, SNS, CloudWatch), Ansible, Terraform

Programming/Scripting: Python, Bash, Shell Scripting

Tools & Platforms: Git, GitHub, Linux

Systems & Administration: Linux Administration, System Monitoring, Troubleshooting, Log Analysis

Support & Operations: Cloud Monitoring, Incident Response, Monitoring Alerts, Root Cause Analysis, Documentation, Debugging Workflows

PROJECTS

Serverless EC2 Monitor & Control Dashboard,

Real-Time EC2 Instance Monitoring and Automation using AWS Lambda, API Gateway, DynamoDB, SES, and S3

- Built a serverless EC2 monitoring and automation system using AWS Lambda, API Gateway, DynamoDB, and S3.
- Automated EC2 instance lifecycle management using usage-based triggers, reducing manual monitoring and shutdown effort by ~1 hour daily.
- Configured AWS-based monitoring, alerting, and scalable infrastructure workflows to improve observability, reliability, response time, and operational availability.
- Documented deployment architecture, workflows, and operational processes.

Zero Downtime Application Deployment using Kubernetes & AWS

- Designed and implemented a highly available cloud-native deployment architecture using Amazon EKS and AWS services.
- Implemented Kubernetes rolling updates, readiness probes, and liveness probes to achieve zero-downtime deployments and self-healing recovery.
- Configured Horizontal Pod Autoscaler (HPA) to dynamically scale workloads based on CPU utilization during high traffic conditions.
- Implemented warm standby disaster recovery architecture using Route 53 health checks and multi-region failover.
- Integrated monitoring and observability tools including Prometheus, Grafana, and CloudWatch for tracking system health and scaling behavior.

VOLUNTEER EXPERIENCE

Android Open Source and Custom ROM Communities,

06/2017 – Present

Open Source Community Volunteer Contributor

- Assisted in testing, validation, and troubleshooting of Android custom operating system builds across multiple devices.
- Investigated system-level issues, crashes, and performance bottlenecks using diagnostic logs and debugging tools.
- Collaborated with distributed developer communities to reproduce bugs, improve system stability, and support issue resolution workflows.
- Worked with recovery, flashing, and restoration utilities while documenting testing observations and troubleshooting steps.

EDUCATION

Masters in Computer Application (MCA), Jain University

2024 – Present | Bengaluru, Karnataka

Specialization in Storage & Cloud Technology (Major), AI & ML (Minor)

Bachelors in Computer Application (BCA), Chandigarh University

2021 – 2024 | Mohali, Punjab

CERTIFICATES

Ethical Hacking from National Institute of Electronics & Information Technology

National Institute of Electronics & Information Technology (NIELIT), Gorakhpur (2021)

Introduction to Industry 4.0 and Industrial Internet of Things

National Institute of Electronics & Information Technology (NIELIT), IIT Kharagpur (2022)

PUBLICATIONS

Published Research Paper: Build a Serverless EC2 Monitor & Control Dashboard with AWS Lambda, API Gateway, and S3,

International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET)